Reinforcement Learning for a Smart Thermostat Energy Saver

CSCE 642 — Reinforcement Learning course project Jessica Salguero and Omar Khan

Summary of the current project

This project studies whether a tabular reinforcement-learning (RL) controller can anticipate occupancy and time-of-use electricity prices in a simulated single-zone home, and how the learned behavior changes as the energy-cost and comfort weights change. The core comparison is an epsilon-greedy Q-learning controller against a reactive deadband thermostat and a random-action sanity-check policy. The initial simulator is a deliberately transparent, first-order resistor-capacitor (RC) model implemented as a Gymnasium environment. It is an educational, controlled experiment, not a calibrated building model or a deployment claim.

The hourly observation contains indoor and outdoor temperature, hour, occupancy, weekday/weekend, current HVAC mode, and a time-of-use price tier. The five actions are off, low heat, high heat, low cool, and high cool. Heating and cooling are represented by signed thermal powers. The reward is the negative weighted sum of interval electricity cost, occupied comfort violation in degree-hours, and a mode-change indicator. The current configuration uses a one-hour time step, `R = 2 deg F/kW`, `C = 5 kWh/deg F`, an occupied comfort band of 68–76 deg F, a deterministic daily outdoor temperature profile, weekday work-hour absence, and an afternoon/evening peak price. These are stated assumptions for reproducibility, not measurements or tuned results; the project guide records their definitions and units. For this initial model, electrical input equals absolute HVAC thermal power (coefficient of performance 1), a simplifying assumption rather than a realistic HVAC efficiency model.

The research question is whether a small tabular policy learns useful pre-conditioning behavior from these signals, and how that behavior trades comfort against cost. We will measure total energy cost, occupied comfort degree-hours, percentage of occupied time in band, HVAC mode changes, and the lower comfort-bound deficit at the first occupied step after an away period. The held-out controller comparison must use the same weather, schedules, episode lengths, and seeds. No performance result is claimed before those experiments are run.

Decisions since the proposal

The proposal left the hour representation, thermal units, and exact occupancy/price schedules open. The implementation uses the exact hour (0–23) and documents the corresponding Fahrenheit, kW, kWh, and hour-based RC units. It fixes weekday absence to 08:00–18:00, keeps weekends occupied, and sets the daily peak-price interval to 16:00–21:00. It also makes electrical input equal to thermal HVAC power (COP 1) so that energy and cost are computable without adding an unsupported equipment-efficiency model. The first experiment remains tabular Q-learning with deadband and random baselines; deep or continuous controllers remain stretch goals. Comfort is reported both as degree-hours and occupied time in band, and the planned arrival metric is the lower-bound temperature deficit rather than an unspecified “comfort” score.

Related work

RL foundations and algorithm choice

Watkins and Dayan established the tabular Q-learning update and its convergence conditions: all relevant state-action pairs must be visited repeatedly and the step sizes must satisfy suitable conditions [1]. Those assumptions make state coverage an important practical limitation here, not a guarantee that a finite training run will learn an effective thermostat. Sutton and Barto provide the practical RL treatment underlying our epsilon-greedy policy and reward formulation [2]. Our first agent uses a table because the action set is small and the resulting policy is inspectable; function approximation is not needed to answer the initial research question.

Mnih et al.'s deep Q-network (DQN) uses replay and a target network to stabilize value learning with function approximation [3]. It is a useful possible extension if the discrete table proves too sparse, but it introduces more training choices than the core comparison requires. Lillicrap et al.'s deep deterministic policy gradient (DDPG) method learns continuous actions with an actor and critic [4], while Schulman et al.'s proximal policy optimization (PPO) uses clipped policy updates [5]. Both address a continuous-control setting more directly than the five-action project; neither is part of the initial implementation. This distinction keeps the first result attributable to the reward, state discretization, and simple thermal dynamics rather than to a larger algorithm stack.

RL for buildings and residential thermostats

Vázquez-Canteli and Nagy survey RL for demand response, comparing modeling and algorithm choices across grid-interactive loads [6]. Wang and Hong review RL for building controls and organize studies by state, action, reward, and environment [7]. Together, these reviews motivate an explicit MDP definition and careful separation between simulated evidence and field validation. Our study narrows the domain to one residential zone and a deterministic, reproducible price/occupancy schedule; it does not address deployment, multi-building transfer, or grid-scale coordination.

Barrett and Linder present a residential HVAC architecture pairing Bayesian occupancy prediction with Q-learning [8]. It is the closest methodological comparison: both use tabular value learning for thermostat decisions, but our initial agent receives the scheduled occupancy state directly and is compared with a deadband controller under a synthetic time-of-use tariff. The proposed experiment can therefore test whether anticipation emerges from the learned policy under known schedules; it does not reproduce their occupancy prediction method.

Ruelens et al. apply batch fitted Q-iteration to thermostatically controlled loads for residential demand response [9]. Their offline regression over collected transitions and forecast-aware policy differ from our online tabular updates and small deterministic simulator. Their work makes the price-responsive thermal-control question relevant, while the implementation contrast clarifies that our project tests accessibility and interpretability, not an alternative claim of algorithmic superiority.

Thermal models and simulation

Bacher and Madsen describe selecting grey-box models for building heat dynamics and estimating physically meaningful thermal parameters from data [10]. Our one-node RC model is a much smaller special case: its resistance and capacitance are hand-set assumptions, not identified from measurements. Its advantage is that the dynamics and time constant are easy to inspect; its limitation is that it omits solar gain, internal loads, humidity, equipment efficiency, and zone coupling.

Crawley et al. describe EnergyPlus as a detailed whole-building energy simulation program [11]. EnergyPlus is better suited to realistic building loads and envelope behavior but is more involved than the inexpensive equation used here. Towers et al. describe Gymnasium's standardized reinforcement-learning environment interface [12]; we use its seeded `reset`/`step` contract and separate termination flags so the environment can be validated independently of the learner. Our

simulator sacrifices fidelity for speed, transparent assumptions, and deterministic test cases.

Synthesis and limitations

Prior work establishes several relevant methods: tabular and deep value learning, policy-gradient control, batch learning for thermostatic loads, and physics-based building simulation [1–12]. The contribution here is narrower: a small inspectable environment that puts occupancy, time-varying prices, comfort, and switching in one reproducible experiment, followed by a direct test of whether an inspectable Q-table learns anticipatory actions. It is not a new RL algorithm and does not claim measured energy savings or real-home validity.

Several limitations must remain explicit in the final report. The synthetic weather and occupancy schedules are known in advance; the RC parameters and HVAC powers are illustrative rather than calibrated; one-hour transitions and linear heat flow omit important building behavior; the state discretization can limit Q-table coverage; and chosen reward weights affect the policy. Results will be reported across held-out seeds with variability and with negative outcomes included. A high-fidelity simulator, uncertain occupancy, continuous control, and transfer between buildings are possible future work, not evidence available from this project.

References

1. Watkins, C. J. C. H., and Dayan, P. “Q-learning.” *Machine Learning*,

8(3–4), 279–292, 1992. https://doi.org/10.1007/BF00992698.

2. Sutton, R. S., and Barto, A. G. *Reinforcement Learning: An Introduction*,

2nd ed. MIT Press, 2018. http://incompleteideas.net/book/the-book-2nd.html.

3. Mnih, V., Kavukcuoglu, K., Silver, D., et al. “Human-level control through

deep reinforcement learning.” *Nature*, 518, 529–533, 2015. https://doi.org/10.1038/nature14236.

4. Lillicrap, T. P., Hunt, J. J., Pritzel, A., et al. “Continuous control with

deep reinforcement learning.” *International Conference on Learning Representations*, 2016. https://arxiv.org/abs/1509.02971.

5. Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O.

“Proximal policy optimization algorithms.” arXiv:1707.06347, 2017. https://arxiv.org/abs/1707.06347.

6. Vázquez-Canteli, J. R., and Nagy, Z. “Reinforcement learning for demand

response: A review of algorithms and modeling techniques.” *Applied Energy*, 235, 1072–1089, 2019. https://doi.org/10.1016/j.apenergy.2018.11.002.

7. Wang, Z., and Hong, T. “Reinforcement learning for building controls: The

opportunities and challenges.” *Applied Energy*, 269, article 115036, 2020. https://doi.org/10.1016/j.apenergy.2020.115036.

8. Barrett, E., and Linder, S. “Autonomous HVAC control, a reinforcement

learning approach.” In *Machine Learning and Knowledge Discovery in Databases: European Conference, ECML PKDD 2015*, pp. 3–19. Springer, 2015. https://doi.org/10.1007/978-3-319-23461-8_1.

9. Ruelens, F., Claessens, B. J., Vandael, S., De Schutter, B., Babuška, R.,

and Belmans, R. “Residential demand response of thermostatically controlled loads using batch reinforcement learning.” *IEEE Transactions on Smart Grid*, 8(5), 2149–2159, 2017. https://doi.org/10.1109/TSG.2016.2517211.

10. Bacher, P., and Madsen, H. “Identifying suitable models for the heat

dynamics of buildings.” *Energy and Buildings*, 43(7), 1511–1522, 2011. https://doi.org/10.1016/j.enbuild.2011.02.005.

11. Crawley, D. B., Lawrie, L. K., Winkelmann, F. C., Buhl, W. F., Huang,

Y. J., Pedersen, C. O., Strand, R. K., Liesen, R. J., Fisher, D. E., Witte, M. J., and Glazer, J. “EnergyPlus: Creating a new-generation building energy simulation program.” *Energy and Buildings*, 33(4), 319–331, 2001. https://doi.org/10.1016/S0378-7788(00)00114-6.

12. Towers, M., Kwiatkowski, A., Terry, J., et al. “Gymnasium: A standard

interface for reinforcement learning environments.” arXiv:2407.17032, 2024. https://arxiv.org/abs/2407.17032.